

# ZHE FU

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## EDUCATION & FELLOWSHIPS

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### Stanford University

[Stanford Energy Postdoctoral Fellowship](#) (*Awarded*) | Mentors: [Eric Darve](#) & [Marco Pavone](#)

Sep. 2026 -

Mechanical Engineering | Aeronautics and Astronautics

Institute for Computational & Mathematical Engineering ([ICME](#))

*Granted three years of independent research funding, with flexible duration*

### University of California, Berkeley

Ph.D.(ABD), Transportation Engineering | Advisor: [Alexandre Bayen](#)

Aug. 2020 - Expected May 2026

Minors: Optimization and Networks, Data Science

M.S., Electrical Engineering and Computer Science | Advisors: [Alexandre Bayen](#) & [Claire Tomlin](#)

Aug. 2022 - May 2026

M.S., Transportation Engineering | Advisors: [Alexandre Bayen](#) & [Xiao-Yun Lu](#)

Aug. 2018 - May 2020

*Affiliations:* Berkeley's Artificial Intelligence Research ([BAIR](#)) | Institute of Transportation Studies ([ITS](#))

### Tongji University

B.S., Transportation Engineering | Advisor: [Xiaohong Chen](#) *Awarded National Scholarship (Top 1%)*

Sep. 2014 - July 2018

### Fudan University

Minor in Bachelor of Law (*10 courses, 30 units in total*)

Sep. 2015 - July 2017

## RESEARCH INTERESTS

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**Methodology:** Physics-informed Machine Learning, Control, Optimization, Numerical Analysis, Human-Centered AI

**Applications:** Mixed Autonomy Systems, Distributed Parameter Systems, Automated Vehicles, Advanced Air Mobility

## SELECTED AWARDS & HONORS & GRANTS

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### Awards:

#### Research

- [Rising Star in EECS](#), EECS department at MIT (70 out of 327, Top 21%) Oct. 2025
- [Runner-up Winner](#), 2025 Berkeley Grad Slam (Campus-wide **sole** winner) April 2025
- [NSF CPS Rising Star](#), 2025 CPS Rising Stars Workshop (30 out of 174, Top 17%) Mar. 2025
- [Rising Star in Mechanical Engineering](#), ME department at CMU (Among 30 selected attendees) Oct. 2024
- [IEEE ITSC 2024 Institutional Lead Award - CIRCLES Consortium](#) Sept. 2024
- [First-place Winner](#), 2023 INFORMS Poster Competition (4 out of 200, Top 2%) Oct. 2023

#### Leadership

- [Eno Fellow](#), Eno Center for Transportation (Among 17 selected graduate students nationwide) June 2025
- [Athena Award for Graduate Student Leadership](#), EDGE in Tech (**sole** winner) May 2025
- Dean's Award for Inclusive Excellence, [Osiki Student Leadership Award](#) (Campus-wide **sole** awardee) April 2024

#### Teaching & Mentoring

- [Outstanding Graduate Student Peer Mentor Award](#), UC Berkeley (Among 4 awardees campus-wide) March 2026
- Evergreen Award for Undergraduate Researcher Mentoring, EECS department at UC Berkeley May 2024
- [Outstanding Graduate Student Instructor](#), UC Berkeley (Top 10%) March 2022

### Scholarships:

- CTF Lipman Family Sustainable Transportation Graduate Fellowship, *California Transportation Foundation* Feb. 2026
- Bridgette Beato Leadership Legacy Graduate Scholarship, *WTS International* Feb. 2026
- [Rodney E. Slater Scholarship](#), *Eno Center for Transportation* June 2025
- Asian American Architects and Engineers Foundation ([AAa/e](#)) [Graduate Scholarship \(First Place\)](#), *AAa/e* Oct. 2024
- [ITS California and California Transportation Foundation Scholarship](#), *ITSCA & CTF* Aug. 2023
- Chinese National Scholarship (Top 1%), *Chinese Ministry of Education* Sept. 2017

### Grants:

- [Stanford Energy Postdoctoral Fellowship](#) (3 years independent funding), *Stanford* Jan 2026
- Berkeley's Artificial Intelligence Research Lab ([BAIR](#)) Graduate Grant (\$12k), *BAIR* May 2024
- Hearts to Humanity Eternal ([H2H8](#)) Graduate Research Grant (\$10k), *H2H8* July 2023

## PUBLICATIONS

### Journals (Published & Under review) \* Equal first author, † Corresponding author

- J1. “Supervised and Unsupervised Neural Network Solver for First Order Hyperbolic Nonlinear PDEs.” [\[link\]](#)  
Submitted to *SIAM Journal on Numerical Analysis*.  
**Zhe Fu**\*†, Zakaria Baba\*, Alexandre M. Bayen, Alexi Canesse\*, Maria Laura Delle Monache, Martin Drieux\*, Nathan Lichtlè\*, Zihe Liu, Hossein Nick Zinat Matin\*, Benedetto Piccoli.
- J2. “Energy-Saving Control of Freeway Traffic: Field Experiments and Results.”  
Submitted to *Nature*.  
Sean McQuade, ..., **Zhe Fu**, ..., Jonathan Sprinkle. [All researchers in MegaVanderTest]
- J3. “Safety-Critical Scenario Generation via Adversarial Imitation in Mixed-Autonomy Traffic.”  
Submitted to *IEEE Transactions on Intelligent Transportation Systems*.  
Qing Lyu, **Zhe Fu**†, Alexandre M. Bayen.
- J4. “Validation and Calibration of Energy Models with Real Vehicle Data from Chassis Dynamometer Experiments.” [\[link\]](#)  
Submitted to *Vehicle System Dynamics*.  
Joy Carpio, Sulaiman Almatrudi, Nour Khoudari, **Zhe Fu**, Kenneth Butts, Jonathan Lee, Benjamin Seibold, Alexandre M. Bayen
- J5. “Kernel-based Planning and Imitation Learning Control for Flow Smoothing in Mixed Autonomy Traffic.” [\[link\]](#)  
Accepted at *Transportation Research Part C*, 2024.  
**Zhe Fu**†, Arwa Alanqary, Abdul Rahman Kreidieh, Alexandre M. Bayen.
  - **First-place Winner**, INFORMS Annual Meeting Poster Competition (4 out of 200), 2023.
  - Honorable Mention. OR/MS Tomorrow Mini-poster Competition (PhD Category), 2023.
  - Accepted by 25<sup>th</sup> International Symposium on Transportation and Traffic Theory ([ISTTT25](#)), 2024.
- J6. “Hierarchical Speed Planner for Automated Vehicles: A Framework for Lagrangian Variable Speed Limit in Mixed Autonomy Traffic.” [\[link\]](#)  
Accepted at *IEEE Control Systems Magazine*, 2024.  
Han Wang, **Zhe Fu**, Jonathan Lee, Hossein Nick Zinat Matin, Arwa Alanqary, Daniel Urieli, Sharon Hornstein, Abdul Rahman Kreidieh, Raphael Chekroun, William Barbour, William A Richardson, Dan Work, Benedetto Piccoli, Benjamin Seibold, Jonathan Sprinkle, Alexandre M. Bayen, Maria Laura Delle Monache.
- J7. “Traffic Smoothing via Connected & Automated Vehicles: A Modular, Hierarchical Control Design Deployed in a 100-cav Flow Smoothing Experiment.” [\[link\]](#)  
Accepted at *IEEE Control Systems Magazine*, 2024.  
Jonathan Lee, ..., **Zhe Fu**, ..., Alexandre M. Bayen. [All researchers in CIRCLES consortium]
- J8. “Evaluating Smart Charging Strategies using Real-world Data from Optimized Plug-in Electric Vehicles.” [\[link\]](#)  
Accepted at *Transportation Research Part D*, 2021.  
Sierra I. Spencer, **Zhe Fu**, Elpiniki Apostolaki-Iosifidou, Timothy E. Lipman.
- J9. “Economic and Environmental Benefits for Electricity Grids from Spatiotemporal Optimization of EV Charging.” [\[link\]](#)  
Accepted at *Energies*, 2021.  
Soomin Woo, **Zhe Fu**, Elpiniki Apostolaki-Iosifidou, Timothy E. Lipman.

### Journals (In Preparation)

- J9. From Density PDEs to Velocity Dynamics: Neural Flux Learning for Traffic Forecasting.  
**Zhe Fu**, Benedetto Piccoli, Alexandre M. Bayen.
- J10. Certified Safe Set Learning via Counterexample-Guided Pareto Control Barrier Functions.  
**Zhe Fu**\*, Xiaoyang Cao\*, Jingqi Li\*, Claire Tomlin, Alexandre M. Bayen.

### Refereed Conference Proceedings (Published & Under review) \* Equal first author, † Corresponding author

- C1. “(U)NFV: Supervised and Unsupervised Neural Finite Volume Methods for Solving Hyperbolic PDEs.” [\[link\]](#)  
*The Fourteenth International Conference on Learning Representations (ICLR)*, 2026.  
**Zhe Fu**\*, Nathan Lichtlè\*, Alexi Canesse\*, Hossein N.Z. Matin\*, Maria L.D. Monache, Alexandre M. Bayen.
- C2. “Unsupervised Anomaly Detection in Multi-Agent Trajectory Prediction via Transformer-Based Models.”  
*IEEE Intelligent Vehicles Symposium (IV)*, 2026.  
Qing Lyu, **Zhe Fu**†, Alexandre M. Bayen.

- C3. “Discovering Safety-Critical Driving Scenarios via Prediction-Based Anomaly Detection.” *Transportation Research Board (TRB) Annual Meeting*, 2026.  
Qing Lyu, **Zhe Fu†**, Alexandre M. Bayen.
- C4. “Position and Speed Estimation Using Deep Learning-based KKL Observer in Mixed Autonomy Traffic Systems.” [\[link\]](#)  
*IEEE Conference on Decision and Control (CDC)*, 2025. **[Oral presentation]**  
Yasmine Marani, **Zhe Fu**, Ibrahima N’Doye, Eric Feron, Taous-Meriem Laleg-Kirati, Alexandre M. Bayen.
- C5. “Neural Network-Based Solvers for Hyperbolic Conservation Laws: Supervised vs. Unsupervised Learning, and Applications to Traffic Modeling” [\[link\]](#)  
*2nd International Conference on Neuro-symbolic Systems (NeuS)*, 2025. **[Oral presentation]**  
**Zhe Fu\***, Alexi Canesse\*, Nathan Lichtlè\*, Hossein N.Z. Matin\*, Zihe Liu, Maria L.D. Monache, Alexandre M. Bayen.
- C6. “Pareto Control Barrier Function for Inner Safe Set Maximization Under Input Constraints.” [\[link\]](#)  
*American Control Conference (ACC)*, 2025. **[Oral presentation]**  
Xiaoyang Cao, **Zhe Fu†**, Alexandre M. Bayen.
- C7. “Towards Understanding Worldwide Cross-cultural Differences in Implicit Driving Cues: Review, Comparative Analysis, and Research Roadmap.” [\[link\]](#)  
*International conference on intelligent transportation systems (ITSC)*, 2024. **[Oral presentation]**  
Yongqi Dong, Chang Liu, Yiyun Wang, and **Zhe Fu**.
- C8. “Kernel-based Planning and Imitation Learning Control for Flow Smoothing in Mixed Autonomy Traffic.” [\[link\]](#)  
*25<sup>th</sup> International Symposium on Transportation and Traffic Theory (ISTTT25)*, 2024.  
**Zhe Fu†**, Arwa Alanqary, Abdul Rahman Kreidieh, Alexandre M. Bayen.
- C9. “Hierarchical Reinforcement Learning for Scalable Lagrangian Variable Speed Limit in Mixed Autonomy Traffic.”  
*Transportation Research Board 103th Annual Meeting*, 2024.  
Han Wang, Jonathan W. Lee, **Zhe Fu**, Sharon Hornstein, Daniel Urieli, Maria Laura Delle Monache.
- C10. “Cooperative Driving for Speed Harmonization in Mixed-Traffic Environments.” [\[link\]](#)  
*IEEE Intelligent Vehicles Symposium (IV)*, 2023.  
**Zhe Fu†**, Abdul Rahman Kreidieh, Han Wang, Jonathan W. Lee, Maria Laura Delle Monache, Alexandre M. Bayen.
- C11. “A Feedback Control Strategy for Traffic Flow Harmonization in Mixed-traffic Environments.”  
*Transportation Research Board 102th Annual Meeting*, 2023.  
**Zhe Fu†**, Abdul Rahman Kreidieh, Alexandre M. Bayen.
- C12. “Learning Energy-efficient Driving Behaviors by Imitating Experts.” [\[link\]](#)  
*International conference on intelligent transportation systems (ITSC)*, 2022. **[Oral presentation]**  
Abdul Rahman Kreidieh, **Zhe Fu**, Alexandre M. Bayen.

#### **Refereed Conference Proceeding (In Preparation)**

- C12. Neural Finite Volume Learning and Control for Mixed Autonomy Traffic.  
**Zhe Fu**, Xuanmian He, Alexandre M. Bayen.
- C13. A Systematic Benchmark of Physics-Constrained Neural Models for Hyperbolic PDE Learning in Traffic Systems.  
**Zhe Fu**, Zihe Liu, Alexandre M. Bayen.

#### **Other Articles**

- O1. “Massive CAV Experiment in Nashville Pits Machine Learning against Traffic Jams.” [\[link\]](#)  
*OR/MS Today*, 2023.  
**Zhe Fu**, Kara Manke, Alexandre M. Bayen.
- O2. “Streamlining CAV Test Data Collection and Evaluation in the Hardware-in-the-Loop Environment.” [\[link\]](#)  
*UC-ITS-2020-23*, 2020.  
**Zhe Fu**, Hao Liu, Xiao-Yun Lu.

#### **REFEREE ACTIVITIES**

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##### **Journal Article Reviewer**

- IEEE Open Journal of Intelligent Transportation Systems
- IEEE Transactions on Intelligent Transportation Systems
- IEEE Transactions on Vehicular Technology
- Nature Partner Journals (npj) Discover Artificial Intelligence
- Nature Partner Journals (npj) Sustainable Mobility and Transport
- Scientific Reports
- Transportation Research Record
- Transportation Research Part C: Emerging Technologies

- Transportation Research Part D: Transport and Environment
- Transportation Science

### Conference Paper Reviewer

- 25th International Symposium on Transportation and Traffic Theory (ISTTT25)
- ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS)
- IEEE American Control Conference (ACC)
- IEEE Conference on Decision and Control (CDC)
- IEEE European Control Conference (ECC)
- IEEE International Conference on Intelligent Transportation Systems (ITSC)
- IEEE Intelligent Vehicles Symposium (IV)
- Transportation Research Board (TRB) Annual Meeting
- World Transport Convention

### Proposal Reviewer

- ITS Berkeley proposals. Transportation Equity Review Committee for ITS Berkeley STRP.

## SELECTED RESEARCH PROJECTS

### Physics-Informed Machine Learning for Hyperbolic PDEs

Sep. 2023 - Present

*Berkeley Artificial Intelligence Research (BAIR)*

- Developed a physics-informed deep learning framework for solving hyperbolic conservation laws (e.g., LWR traffic model), combining neural flux approximation with finite volume structures.
- Designed the model to operate on the same stencils as traditional finite volume (FV) methods, matching the computational cost of first-order schemes while enabling easy extension to higher-order accuracy in both space and time.
- Framework supports both supervised and unsupervised training paradigms:
  - Supervised learning uses entropy solutions as ground truth and achieves higher accuracy.
  - Unsupervised learning leverages a weak formulation of the PDE to learn from unlabeled initial conditions.
- Captured entropy solutions reliably under both supervised and unsupervised training modes.
- Outperformed all known first-order FV schemes (e.g., Godunov, Engquist-Osher) across multiple flux models.
- Demonstrated superior accuracy over numerical baselines on real-world traffic datasets, including [PeMS](#), [INRIX](#), [drone-captured highway videos](#), and the [I-24 MOTION dataset](#).
- Established a generalizable and interpretable framework for learning PDE-governed dynamics, with scalability to broader scientific and engineering systems.

### Congestion Impacts Reduction via CAV-in-the-Loop Lagrangian Energy Smoothing

Sep. 2019 - Present

[CIRCLES: World's Largest Mixed Traffic Field Test \(100 AVs\)](#)

*Berkeley Artificial Intelligence Research (BAIR)*

- Constructed a feedback control strategy on Automated Vehicles (AVs) to smooth stop-and-go waves, achieving on average over 15% energy savings within simulations of congested events with only 4% AV penetration.
- Utilized DAgger, an online imitation learning procedure to capture traffic flow harmonization behaviors using only decentralized observations.
  - A Multi-Layer Perceptron (MLP) is used with hidden layers (32, 32, 32), and a ReLU nonlinearity.
  - Simulation results show the success of the approach in matching the expert performance without access to global states.
- Designed the project's core "Speed Planner", a higher-level control module that provides real-time, energy-efficient speed suggestions to AVs to execute.
- Conducted large operational tests with such control strategies deployed on 100 AVs on I-24 highway. Analyzed field data showing 10-20% fuel savings near controlled AVs, despite AVs representing only 1-2% of total traffic.
- Collaborated with many industry partners including [Toyota](#), [Nissan](#) and [General Motors](#).

### Streamlining CAV Test Data Collection and Evaluation in the HIL Environment

Sep. 2019 - May 2020

*California PATH*

- Built a prototype Hardware-in-the-Loop (HIL) tool in C++ to connect Aimsun and MySQL database to facilitate data collection, storage and evaluation capabilities for easier data analysis in HIL tests.
- The tool has been applied in the AVs field experiments to prove tested algorithms' efficiency.

### Evaluating Plug-In EV Using Optimization Framework and Real-world Study Data

May 2019 - Sep. 2019

[BMW Charge Forward Program](#)

*Transportation Sustainability Research Center(TSRC)*

- Established two linear and convex optimization models: Fixed-location and Inter-location models to schedule the charging time and location of Electric Vehicles (EVs).

- Evaluated the optimization models in Python with three different objective functions in terms of energy load, emission, and renewable-energy usage on the user data collected from [BMW Charge Forward Program](#).

## INDUSTRY & INTERNSHIP EXPERIENCE

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**Research Intern**, [Honda Research Institute](#), Ann Arbor, MI, USA May 2023 - Aug. 2023  
Worked on multi-agent robots collision avoidance in social navigation

**Research Intern**, [Tsinghua Berkeley Shenzhen Institute](#), Shenzhen, China July 2018 - Aug. 2018  
Worked with Prof. Yi Zhang on Global Competence and Leadership Survey

## TEACHING & MENTORING EXPERIENCE

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**Graduate Student Instructor**, UC Berkeley Aug. 2021 - Dec. 2021  
CE262 Analysis of Transportation Data (Graduate Level)

- Enrollment: 57
  - PhD/Master **core** course for SYS and TRANS students in CEE department
  - Students' degree programs include Undergraduate, MEng, MS, MBA and PhD
  - Students' departments include CEE, CED and Haas School of Business.
- Ratings: 4.67/5.00 (department average: 4.41, response ratio: 77.19% | 44/57)
  - Helpfulness 4.84/5.00 (avg: 4.57); Communication 4.77/5.00 (avg: 4.48)
  - Inclusive environment 4.79/5.00 (avg: 4.49); Accessibility 4.86/5.00 (avg: 4.67)
- Awarded [Outstanding Graduate Student Instructor](#) (Top 10%)
- Invited 3 times (2023/2024/2025) to serve as one of three panelists at International GSI conference held for around 800 first-time GSIs at UC Berkeley.
- Invited to serve as the panelist for the pedagogy class for new GSIs (CE375)

**Lecturer**, UC Berkeley July 2025  
BAIR-NCKU Summer AI Workshop (Graduate Level)

- Enrollment: 6 (Masters Students from NCKU)
- Served as the main student coordinator for the week-long module on "AI & Transportation".
- Instructed the cohort on "Learning and controls in Mixed Autonomy Traffic Systems" and "Neural Finite Volume Methods".

**Lecturer**, UC Berkeley May 2023  
Girl Scout Engineering Day

- Enrollment: 60 (Elementary/Middle School Students from Girl Scout)
- Led a hands-on programming workshop for over 60 elementary and middle school girls, teaching basic coding principles.
- Mentored and inspired a diverse group of young girls to build confidence and pursue future careers in STEM through an engaging introduction to university-level engineering.

**Volunteer Teacher**, Lianhua Village School Jan. 2016 - Feb. 2016  
Basic Health and Hygiene

- Enrollment: 15 (Elementary School Students in "Hui" minority)
- Volunteered to teach elementary school students at a religious school located in Lianhua village, a less-developed corner of China's mountainous west.
- Shocked by the poor health resources the left-behind children have, I designed the "Basic Health and Hygiene" course on my own, including creating the syllabus, preparing the materials and delivering 10 one-hour lectures.

**Guest Lecturer**, CMU & UCLA April 2026  
CMU 12787 Special Topics: Physics-Informed Machine Learning

- Enrollment:
- Instructed on physics-informed learning and control for mixed traffic systems.

UCLA CEE C181/C281 Traffic Engineering Systems: Operations and Control (Undergrad & Graduate Level) Nov. 2025

- Enrollment: 30 (Undergrads & Graduates)
- Instructed on the theory and application of modeling and control in mixed-autonomy traffic systems.



**Mentor**, UC Berkeley  
Undergraduate Students

CEE Scholars research mentorship program (for undergrads from PREP and T-PREP programs)

Spring 2024

*Sami Seyedjafar Kashi*, CEE, Spring 2024.

BAIR undergraduate mentoring program

Oct. 2023 - Present

*Atharva Gupta*, EECS, 2023 - 2024.

*Nicole Han*, EECS & DS, 2023 - present. Currently software engineer at X.

Mobile Sensing Lab undergraduate mentoring

Oct. 2021 - Present

*Jordan Kam*, Aerospace Engineering, 2025 - present. Incoming Ph.D. student at Caltech.

*Qing Lyu*, exchange student from Tongji U, 2025 - present. Currently M.S. student at UC Berkeley CEE.

*Zakaria Baba*, research intern from École Polytechnique, 2025. Currently Ph.D. student at Caltech CMS.

*Martin Drieux*, research intern from École Polytechnique, 2025. Currently M.S. student at Carnegie Mellon University MSML.

*Alexi Canesse*, research intern from École Normale Supérieure de Lyon, 2025. Currently Ph.D. student at École Polytechnique.

*Xiaoyang Cao*, exchange student from Tsinghua U, 2024 - present. Currently Master student at MIT TPP.

*Jiaying Yang*, exchange student from Tongji U, 2023 - 2024. Currently M.S. student at UC Berkeley CEE.

*Ashwin Dara*, EECS, 2023 - 2024. Currently software engineer at Doordash.

*Eric Cheng*, EECS, 2021 - 2022. Currently software engineer at Unity.

Graduate Students

EECS MEng capstone project mentor

Sep. 2023 - May 2024

*Alvin Bao*, EECS, 2023 - 2024. Currently software engineer at Netflix.

*Tony Luo*, EECS, 2023 - 2024. Currently software engineer at Doordash.

*Abirami Sabbani*, EECS, 2023 - 2024. Currently software engineer at Walmart Global Tech.

*Cathy Wang*, EECS, 2023 - 2024. Currently software engineer at AWS.

TRANSOC graduate student mentoring program

Oct. 2020 - Present

*Xuanmian He*, CEE, 2025 - present. Currently M.S. student at UC Berkeley.

*Zihe Liu*, CEE, 2024 - present. Currently Ph.D. student at UC Berkeley.

*Juanwu Lu*, CEE, 2021 - 2022. Currently Ph.D. student at Purdue University.

*Jianxuan Yin*, CEE, 2020 - 2021. Currently construction technology engineer at XL Construction.

**INVITED TALK & PRESENTATIONS**

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- Poster presentation, **ICLR 2026**, Rio de Janeiro, Brazil, April 2026
- Poster presentation, **AI & PDE workshop at ICLR 2026**, Rio de Janeiro, Brazil, April 2026
- Oral presentation, **NorCal Control Workshop**, Monterey Bay, CA, April 2026 ([link](#))
- Seminar talk, **ASU**, Tempe, AZ, April 2026
- Guest lecturer, **CMU**, Pittsburgh, PA, April 2026
- Lightning talk, **COTA TRB Winter Symposium**, Washington DC, Jan. 2026
- Poster presentation, **TRB Annual Meeting 2026**, Washington DC, Jan. 2026
- Poster presentation, **NeurIPS 2025**, San Diego, CA, Dec. 2025
- Spotlight talk, **BARS 2025**, Stanford, CA, Nov. 2025
- Panelist, **Agents, Memory, and Simulation**, Google & SESE, Berkeley, CA, Nov. 2025 ([link](#))
- Guest lecturer & Seminar talk, **UCLA Mobility Lab**, Los Angeles, CA, Nov. 2025
- Poster presentation, **MIT EECS Rising Stars Workshop**, MIT, Boston, MA, Oct. 2025 ([link](#))
- Job Market Showcase talk, **INFORMS Annual Meeting 2025**, Atlanta, GA, Oct. 2025 ([link](#))
- Seminar talk, **USC AAI Seminar**, USC, Los Angeles, CA, Oct. 2025
- Seminar talk, **eMERGE Seminar**, UC Berkeley, Berkeley, CA, Sept. 2025 ([link](#))
- Oral presentation, **ACC 2025**, Denver, CO, July 2025
- Invited poster presentation, **ACC 2025 Workshop W05**, Denver, CO, July 2025 ([link](#))
- Oral presentation, **NeuS 2025**, Philadelphia, PA, May 2025
- Award remark, **EDGE in Tech Athena Award Ceremony**, Berkeley, CA, May 2025  
“Stay Safe, Stay Humble – But Speak Up” , ([link](#))
- Invited talk, **Mobility-X Lab**, Rice U, Houston, TX, May 2025

- Invited talk, **JTL Urban Mobility Lab**, MIT, Boston, MA, April 2025 ([link](#))
- Seminar talk, **Harvard EE Seminar**, Harvard, Boston, MA, April 2025 ([link](#))
- Invited talk, **Zardini Lab**, MIT, Boston, MA, April 2025
- Grad Slam Competition, **Berkeley Grad Slam**, UC Berkeley, Berkeley, CA, April 2025  
“Stop-and-Go No More: How a Few Smart Cars Can Fix Traffic Jams” ([link](#)), **Runner-up Winner Campus-wide**
- Poster talk, **NSF CPS Rising Stars Workshop 2025**, Nashville, TN, March 2025
- Session talk, **POMSHK 2025**, Hong Kong, China, Jan. 2025
- Session talk, **HKSTS 2024**, Hong Kong, China, Dec. 2024
- Spotlight talk, **BARS 2024**, UC Berkeley, Oct. 2024
- Poster talk, **ISTTT25**, Ann Arbor, MI, July 2024
- Invited talk, **INFORMS Annual Meeting 2023**, Phoenix, AZ, Oct. 2023
- Poster competition, **INFORMS Annual Meeting 2023**, Phoenix, AZ, Oct. 2023  
“Cooperative Connected Automated Vehicle Control: Strategies For Speed Harmonization In Mixed Autonomy Traffic.”, **First Place Winner of Multidisciplinary Application Track**
- Interactive poster presentation, **IEEE IV 2023**, Anchorage, AK, June 2023
- Invited talk, **IEEE IV Workshop**, Anchorage, AK, June 2023
- Poster presentation, **Berkeley Deep Drive Symposium**, March 2023
- Poster presentation, **TRB Annual Meeting 2023**, Washington DC, Jan. 2023
- Spotlight speak & Poster presentation, **BARS 2022**, UC Berkeley, Nov. 2022
- Poster presentation, **ITS 75 Celebration**, UC Berkeley, Oct. 2022

## MEDIA & PRESS

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- **CIRCLES work featured in Press**, Fortune ([12/22](#), [11/22](#)); AP News ([11/22](#)); Popular Science ([11/22](#)); PR Newswire ([11/22](#)); TechXplore ([11/22](#)); UC Berkeley News ([11/22](#)); Vanderbilt University News ([11/22](#)); Fox Channel 2 KTVU ([11/22](#)); Fox Channel 17 WZTV Nashville ([11/22](#)); ABC 7 News ([11/22](#)); WKRN News Channel 2 ([11/22](#), [11/22](#)); WABI News Channel 5 ([11/22](#)); WTVF News Channel 5 ([11/22](#)); WSMV Channel 4 ([11/22](#)); Main Street Media of Tennessee ([11/22](#)); Motor ([11/22](#))
- **Outstanding Graduate Peer Mentor Award**, [Website Announcement](#)
- **CTF Lipman Family Sustainable Transportation Graduate Fellowship**, ITS Berkeley ([3/26](#))
- **Stanford Energy Postdoctoral Fellowship**, Stanford Energy ([2/26](#))
- **2025 MIT EECS Rising Star**, [Website Announcement](#)
- **2025 ENO Fellow**, Eno Center for Transportation ([6/25](#), [5/25](#)); UC Berkeley CEE ([6/25](#)); ITS Berkeley ([4/25](#))
- **Athena Award for Graduate Student Leadership**, [Website Announcement](#); CITRIS ([5/25](#)); ITS Berkeley ([5/25](#))
- **2nd Place Winner of 2025 Berkeley Grad Slam**, UC Berkeley Graduate Division ([4/25](#)); ITS Berkeley ([4/25](#)); UC Berkeley CEE ([6/25](#))
- **NSF CPS Rising Star**, [Website Announcement](#)
- **CMU ME Rising Star**, UC Berkeley CEE ([10/24](#)); ITS Berkeley ([10/24](#))
- **AAa/e Scholarship**, UC Berkeley CEE ([10/24](#)); ITS Berkeley ([10/24](#))
- **IEEE ITSC 2024 Institutional Lead Award**, UC Berkeley CEE ([10/24](#)); UC Berkeley EECS ([8/24](#)); ITS Berkeley ([9/24](#))
- **Dean’s Award for Inclusive Excellence**(Osaki Student Leadership Award), UC Berkeley Division of Student Affairs ([4/24](#)); UC Berkeley CEE ([4/24](#))
- **Honorable Mention of 2023 OR/MS Tomorrow Mini-Poster Competition**, [Website Announcement](#)
- **1st Place Winner of 2023 INFORMS Best Poster Competition**, UC Berkeley CEE ([2/24](#)); ITS Berkeley ([1/24](#))
- **ITSCA & CTF Scholarship**, UC Berkeley CEE ([8/23](#)); ITS Berkeley ([8/23](#))
- **H2H8 Research Grant**, ITS Berkeley ([8/23](#))
- **Outstanding GSI**, [Website Announcement](#); UC Berkeley CEE ([4/22](#)); ITS Berkeley ([5/22](#))

## LEADERSHIP & SERVICE

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### Leadership & Outreach

- Junior Researcher Engagement Officer, [REproducible Research In Transportation Engineering \(RERITE\) Working Group](#) July 2025 - Present
- Student Lead Coordinator, BAIR-NCKU Summer AI Workshop July 2025
- Initiator, [IEEE Technical Committee: Automated Mobility in Mixed Traffic](#) Mar. 2025 - Present
- Organizer, ITSC 2024 & 2025 Workshop “[Automated Mobility in Emerging Mixed Traffic](#)”, IEEE ITSS 2024 & 2025
- Sole Graduate Representative in CEE DEIB Committee, UC Berkeley Aug. 2023 - July 2024
- **Founder** of Representation of Asian and Pacific Islander in the Departement of Civil and Environmental Engineering ([RAPID-CEE](#)), UC Berkeley
- President of [RAPID-CEE](#), UC Berkeley May 2023 - Aug. 2024
- Graduate Assembly Delegate, UC Berkeley Aug. 2022 - Aug. 2024
- Funding Committee Member of Graduate Assembly, UC Berkeley Aug. 2022 - Aug. 2024
- Executive Board Member of Women in Computer Science Engineering ([WiCSE](#)), UC Berkeley Aug. 2022 - Aug. 2023
- Vice president of the Student Union, Transportation Engineering School, Tongji University July 2016 - July 2017
- Organizer, the **first** Asian and Pacific Islander (API) Awareness event in CEE department. Feb. 2023
- Organizer, the **first** API Social event in CEE department. May 2023
- Organizer, Berkeley-Stanford Meetup for Women in EECS. April 2023
- Volunteer, Girl Scouts Engineering Day. May 2023

### Professional & Campuswide Service, UC Berkeley CEE & EECS

- Panelist, Path to the Professoriate ([P2P](#)) Program, UC Berkeley Nov. 2025
- PhD Coordinator, [ITS Berkeley Seminar Series](#) Sep. 2024 - May 2025
- Panelist, International Graduate Student Instructor (GSI) conference, UC Berkeley Aug. 2023 & 2024 & 2025
- Core member, CEE Faculty Search Student Committee 2023
- Member, EE Faculty Search Student Committee 2023
- Member, CS Faculty Search Student Committee 2023
- Volunteer, admitted students visit days in EECS and CEE. 2020 - Present
- Volunteer, TU Delft Department of Transport and Planning Visit. April 2023
- Panelist, pedagogy class for new GSIs in CEE (CE375) April 2022

### Professional Affiliations

- Student Member, #4617381, Association for Computing Machinery (ACM)
- Student Member, #12374061, American Society of Civil Engineers (ASCE)
- Student Member, #108, Chinese Overseas Transportation Association (COTA)
- Member, Graduate Women of Engineering (GWE)
- Student Member, #99088722, Institute of Electrical and Electronics Engineers (IEEE)
- Student Member, #1938374, Institute for Operations Research and the Management Sciences (INFORMS)
- Student Member, #46375, Production and Operations Management Society (POMS)
- Student Member, #021030965, Society for Industrial and Applied Mathematics (SIAM)
- Member, Society of Women Engineers (SWE)
- Student Member, Women's Transportation Seminar (WTS)

## OTHER AWARDS & HONORS

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### Other Awards:

- Honorable Mention, 2023 OR/MS Tomorrow Mini-poster Competition (PhD Category) Nov. 2023
- **Best Design Presentation**, 2018 BMW Next Mobility Youth Camp Aug. 2018
- Second Prize, 12th National Transportation Technology Competition (Team Leader, Rank: 9/302) May 2017
- **First Prize**, Transportation Technology Competition at Tongji University (Team Leader, Rank: 2/59) April 2017
- Meritorious Winner, US Mathematical Contest in Modeling (Team Leader, Top 7%) April 2017
- Second Prize, National English Competition for College Students May 2017
- Second Prize, China Mathematical Olympiad (Provincial Level) Oct. 2013
- Second Prize, China Physics Olympiad (Provincial Level) Oct. 2013

### Other Scholarships:

- Professional Development Award, *UC Berkeley* Aug. 2024 & 2025



- Department Award - EECS, *UC Berkeley* 2023
- Block Grant Award - CEE, *UC Berkeley* 2019 - 2023
- Summer Department Award - ITS, *UC Berkeley* May 2021
- Runner-up “Tingya” Outstanding Student Scholarship Competition, *Tongji University* May 2017

## REFERENCES

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- Dr. Alexandre M. Bayen**, Liao Cho Professor (Thesis Advisor) **Email: [bayen@berkeley.edu](mailto:bayen@berkeley.edu)**  
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*Director - GSI Teaching and Resource Center*  
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